Isabella Herrera Pilonieta
April 9th, 2026

A. Vibe Coding (30 points)

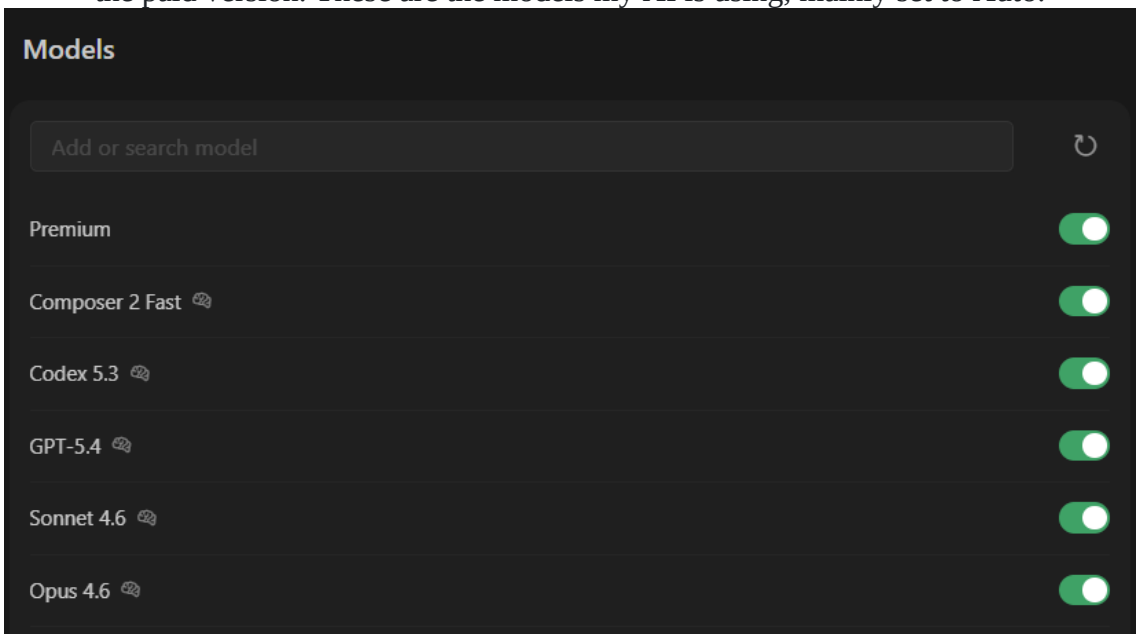
Topic: Add graphics to the Minesweeper knowledge example

1. **Add a paragraph (or more) describing the improvement(s) you made.**

The improvements I made to the Minesweeper knowledge examples can be summarized as follows: I introduced a live, playable version of the game to make the experience more interactive and engaging. I also added a visual panel that displays the game's knowledge base in real time, helping users better understand how decisions are made as the game progresses. To support this, I included clear instructions for interpreting the information in the knowledge panel. Finally, I refined the game's overall formatting and design to create a more user-friendly, visually appealing experience.

2. **Clearly identify the AI system you are using, including version, i.e., ChatGPT 4.5 thinking and paid vs. Free.**

For this homework, I use the Cursor application to write my code. Cursor includes an agent that lets me send files and instructions to help me with the code. The version I am using is the paid version. These are the models my AI is using, mainly set to Auto.



3. **Add a paragraph (or more) identify any issues or problems you encountered.**
4. **Keep a running list of the prompts you used, number them, and add them to your comments.**

Prompt #1

"Okay, the idea of the project is the following.
The main goal: Add graphics to the Minesweeper knowledge example
But we are going to create the game, and as we open cells and select mines, we are going to show the logic knowledge as a visual as well. So, we will have the minesweeper itself, and the knowledge of the state of the board as we go through the game.

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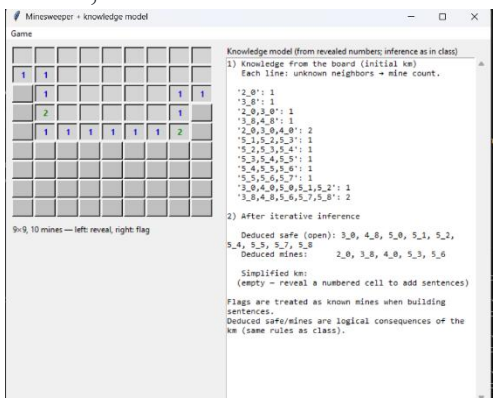
We will use python, logic models, and tinker to show the GUI”

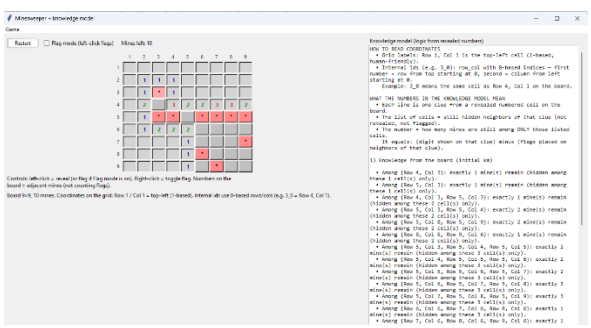


Issue

In the first prompt, I gave the AI the code that implements the Minesweeper gameplay and its logic; however, it is hard to understand what the numbers mean and how the code reads rows and columns. The text is hard to read, and it is unclear whether the instruction ends or the data follows. There are not buttons to restart the game or add flags.

Prompt #2

Now, we need to make it human readable. The code should tell us the number of column and row, and if it’s already there what does the number mean? It says 3_0 are we starting to count from 0 or 1? The game also needs to have all the features as if I was playing, restart button and flag option.



Issue

In the second prompt, I gave AI the issue with the instructions and the knowledge process. The information is difficult to understand, and you cannot distinguish between the knowledge base, where it is safe, and the mines.

Prompt #3

The improvement has been good but the code is still hard to understand. Let’s organize the GUI following these instructions:

1. The instructions of the knowledge base and what does the number mean can be underneath the game. That means the how to read coordinates until “what the numbers of the knowledge model mean” This instruction can be explain in a static place NOT THE SAME AS THE KNOLEDGE BASE DIAGRAM.

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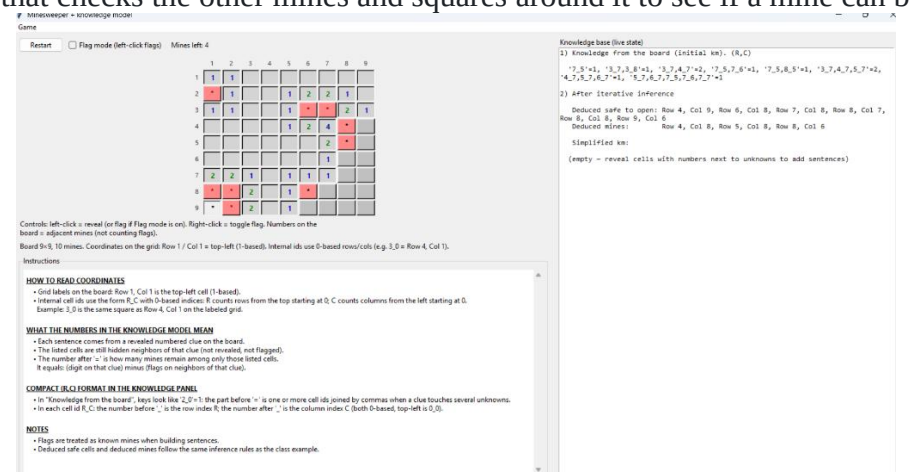
2. In the knowledge from the board (initial km) change the format to ‘2_0’=1, ‘3_8’=0 etc like we had in the last version. However, in the instruction add the instruction of which number is the row and which one is the column. So “knowledge from board initial km. (R,C).
3. Do NOT CHANGE “after iterate reference
4. Add bold and underlines format for title to identify the difference in instructions.}
5. Add the “notes” in the instructions from instruction #1

Issue

For the third prompt, one issue was that in the bottom corner, column 1, row 9, we can see a mine; however, there is no way to see that the cell had a mine. It shouldn’t have a mine, because the grid ends in that square. There is a logic issue when putting the mines. I need to add a rule that checks the other mines and squares around it to see if a mine can be there.



Prompt #4

Making progress, let do these instructions now:

1. There is a log issue when putting the mines. In the last example column 1, row 9 there was a mine, however column 1, row 8, column 2, row 8, and column 2, row 9 are mines, how is possible that column 1, row 9 is a mine if there are no number to see if a mine is there.
2. The text that talks about the controls and board 9x9 should also be inside the instructions box.
3. The knowledge base information is fine but every knowledge should be the as row, not all of the knowledge in a line.
4. Make the knowledge base live information bold. Just the instructions 1 and 2. And inside the instruction 2 make deduced safe and deduced mines bold.
5. There is also an option to change the game from beginner to expert but there is no button to change the mode, add this as well

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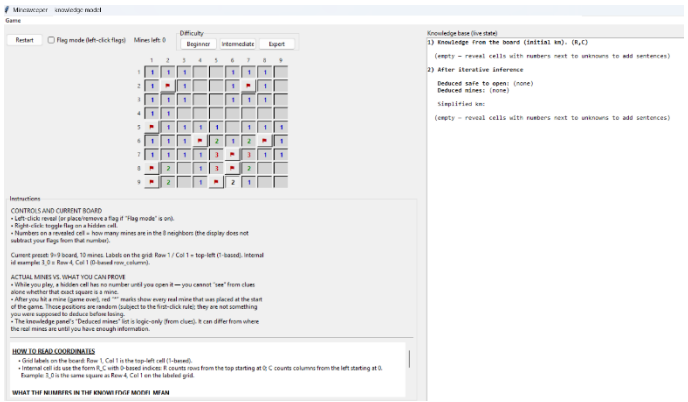

Issue
In this prompt, the issue is that when you click a button without a mine, you cannot see the number or whether it is empty until you click the next one. Another issue is that the instructions should ensure the 'how to read' box is not divided as it appears.

Prompt #5
Last couple changes are:

1. The instructions and how to read coordinates should be inside. But I don't need the ACTUAL MINES VS WHAT YOU CAN PROVE.
2. Let's make it pretty, not so boring. Make some design magic in the minesweeper.

Issue
```
Traceback (most recent call last):
  File "c:\Users\TALLER-TECNOLIDER\OneDrive\Personal\COLLEGE\HU Spring 2026\AI and Machine Learning\p7\minesweeper_gui.py", line 560, in <module>
    MinesweeperApp().run()
    ^^^^^^^^^^^^^^^^^^^
  File "c:\Users\TALLER-TECNOLIDER\OneDrive\Personal\COLLEGE\HU Spring 2026\AI and Machine Learning\p7\minesweeper_gui.py", line 76, in __init__
    self._build_layout()
  File "c:\Users\TALLER-TECNOLIDER\OneDrive\Personal\COLLEGE\HU Spring 2026\AI and Machine Learning\p7\minesweeper_gui.py", line 309, in _build_layout
    kb_header = tk.Frame(right, bg=BG_PANEL, padx=0, pady=(0, 6))
                ^^^^^^^^^^^^^^^^^^^
```

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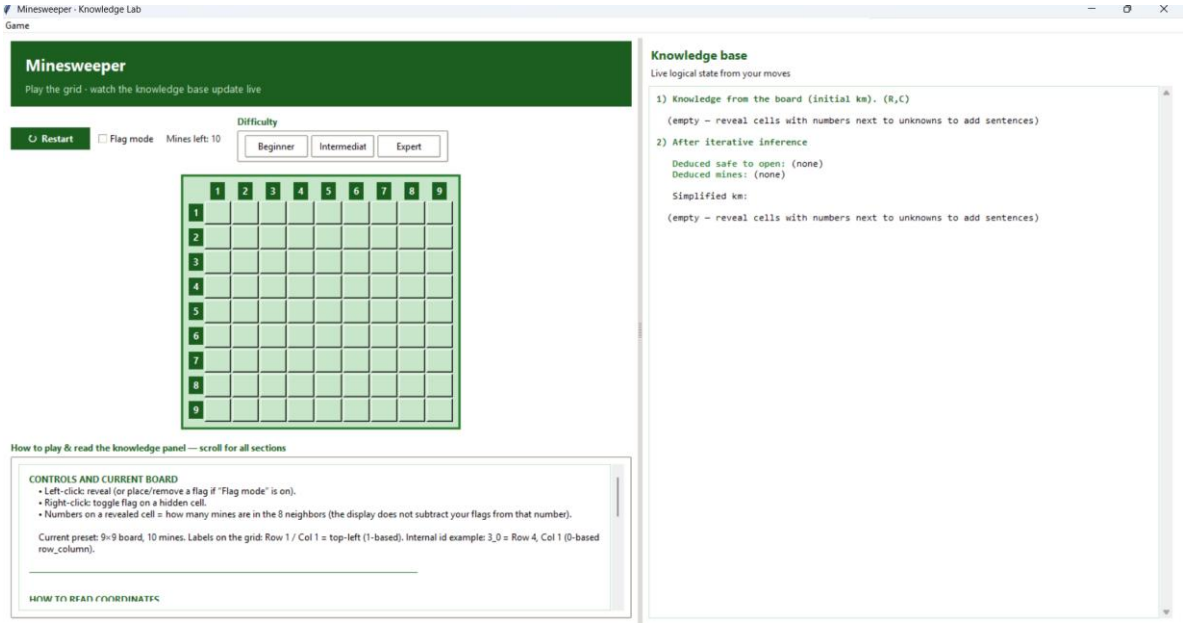
what is wrong with the code and what is not working

Prompt #7
When we change the design in the box that is “how to play and current board” there is no way to move the instruction box to see what is next. Is doesn’t move so I cannot see the rest of the instructions. And use the colors white, black, and forester green for the design.



B. Images (15 points)

1. **Clearly identify the AI system you are using, including version**

Prompt # 1
- Deepai.org

Prompt #2
- Canva AI

Prompt #3
- Deepai.org

Prompt #4
- Canva AI

2. **List the prompt(s) you used**

Prompt #1
I want you to create an image of the game Minesweeper. In this game there are two boxes the left box has in the top the title "Minesweeper" with a small sub header "play the grid and watch the knowledge" underneath the header had button of restart, flag mode, mines left, difficulty and three buttons beginner, intermediate, expert. Underneath this is the game 9x9 board and underneath the game is a box that carries the how to play & read the knowledge panel -- scroll for all sections. Now in the right box the title is knowledge base. With instructions there are three instructions

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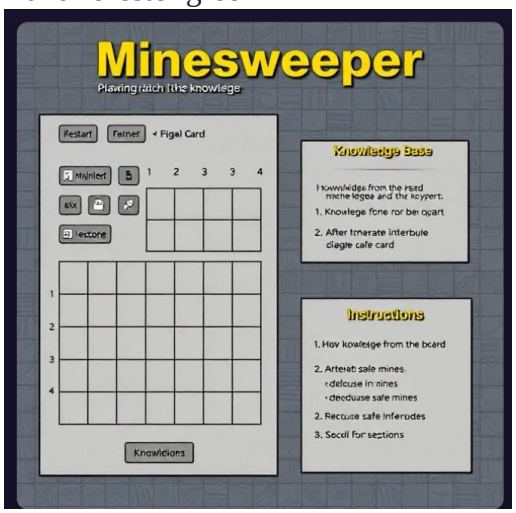
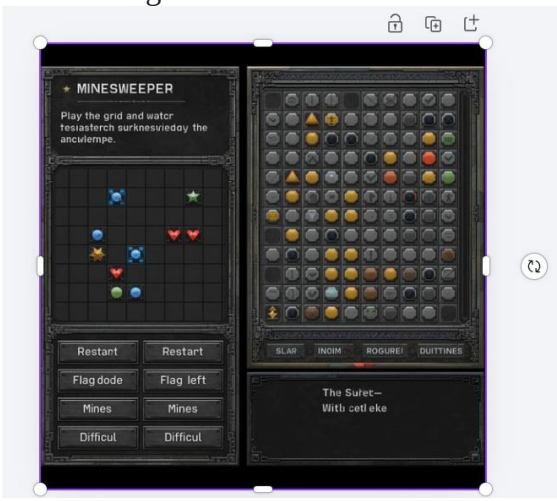
1. Knowledge from the board.
2. After intenerate inference
in this includes deduce safe mines, deduce the mines
Use the colors white, green and forester green



Prompt #2
Please create an image of the game Minesweeper. In this game, there are two boxes. The left box has, at the top, the title "Minesweeper" with a small subheader "play the grid and watch the knowledge." Underneath the headerwered buttons restart, flag mode, mines left, difficulty, and three buttons beginner, intermediate, and expert.
Underneath this is the 9x9 game board, and underneath the game is a box that contains the how to play & read the knowledge panel -- scroll for all sections.
Now, in the right box, the title is knowledge base. instructions there are three instructions
1. Knowledge from the board.
2. After the intenerate inference
, this includes deducing safe mines, deducing the mines
Use the colors white green forester green.



Prompt #3

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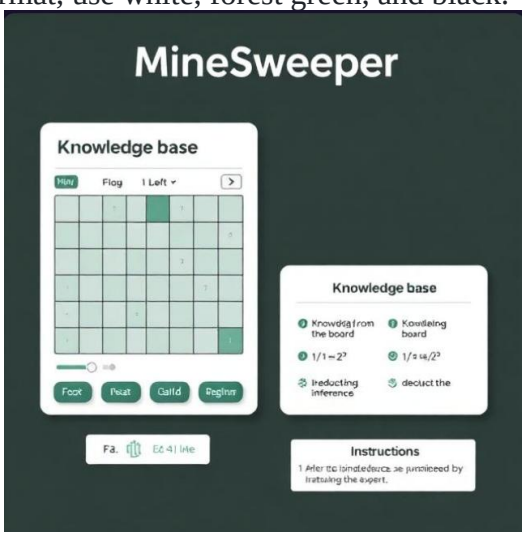
Please create an image of the game Minesweeper. In this game, there are two divisions of boxes. One in the left corner that includes, at the top, the title "Minesweeper" with a small subheader "play the grid and watch the knowledge." Under the header, include buttons for each option: restart, flag mode, mines left, difficulty, beginner, intermediate, and expert. After the header title and buttons, add the minesweeper game grid, a 9x9 board. Underneath the game, add a box that includes how to play the game & how to read the knowledge panel.
Now, in the right corner box, the title for this box is “knowledge base. instructions there are three instructions.” Inside this box, there are three titles for instructions
1. Knowledge from the board.
For this one add numbers like “1_0=2” or “1_5, 1_9=2”
2. After the intuitive inference
After the instruction title, add two subtitles: "deducing safe mines" and "deducing the mines".
For the game's general format, use white, forest green, and black.



Prompt #4
Please create an image of the game Minesweeper. In this game, there are two divisions of boxes. One in the left corner that includes, at the top, the title "Minesweeper" with a small subheader "play the grid and watch the knowledge." Under the header, include buttons for each option: restart, flag mode, mines left, difficulty, beginner, intermediate, and expert. After the header title and buttons, add the minesweeper game grid, a 9x9 board. Underneath the game, add a box that includes how to play the game & how to read the knowledge panel.
Now, in the right corner box, the title for this box is “knowledge base. instructions there are three instructions.” Inside this box, there are three titles for instructions
1. Knowledge from the board.
For this one add numbers like “1_0=2” or “1_5, 1_9=2”
2. After the intuitive inference
After the instruction title, add two subtitles: "deducing safe mines" and "deducing the mines".
For the game's general format, use white, forest green, and black.

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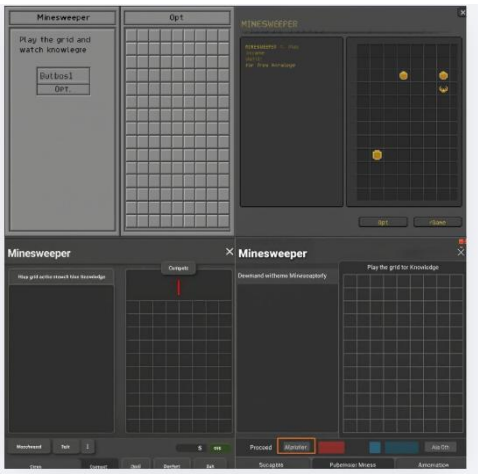

I asked AI to generate more options based on the similar ones using the same prompt. These were the best 4 options I got

3. **Describe any issues or problems you encountered**

Some of the issues I got from the image AI were related to the layout and the way the grid is divided for the game.

With DeepAI.org, it was easier to get close to the actual layouts. At the same time, the AI understood the idea of changing the colors and using them to format the game. The code does not organize the ideas correctly; even though I specify "underneath the board game," the AI still puts some stuff beside the game. One of my thoughts was that I needed to add more detail and dimensions so the machine would not confuse left, right, up, or down in every situation.

With Canva, the application is behind how it should be. Canva did not understand most of the prompt. In the first, it did not take into account the colors; it created two games side by side and duplicate buttons that are not supposed to be there. In the last prompt, where I tried to create various types, the AI kept understanding the left and right as opposite. And again, the AI ignores the color formatting for the game, at the same time, I do not think Canva understood what " minesweeper " is

Overall, some of the images are my fault due to insufficient detail; however, the AI did not identify the dimensions as it should have.

C. Notebook LM (15 points)

1. **List the prompt(s) you used**

Prompt #1

I need to create a presentation that show the improvements that I made to a code of minesweeper using the algorithms behind minesweeper and the game itself.
The start of the project was just the logic of the game, if we open a cell which cells are safe and which have mines. My improvements in this code were create the game live, so a person can play it and beside the game they can see how the logic works for each movement they make. Using the minesweeper algorithm, the player can see which mines he can open and which ones he can flag. I also added design features to make the code better.

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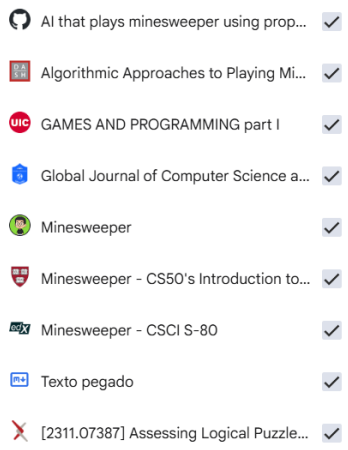
Prompt #2
“ Ya subí mis archivos para hacer la presentación” → Forgot to type in English “I already submitted my files for the presentation”

Prompt #3
Amazing presentation, but I need it to be all in English

Prompt #4
There are two pasted texts that I added: the short one is the old version of Minesweeper that I had, and the other is the new version, which shows the game board and the live knowledge base while the player is playing. Create a presentation that highlights the improvements and changes made in the new code.

2. **List the resources you uploaded**

My list of resources is from the Harvard course Minesweeper and the knowledge topic. The “Texto pegado,” which means text copied, is the whole minesweeper_gui.py



3. **Describe any issues or problems you encountered**

My first issue was that the entire presentation was in Spanish, even though all my prompts were in English.

On slide #6, in the second circle, there is a word that cannot be understood.

On the third try I did the presentation I could see less quality in the slides. Also slide #6 just have the alphabet in upper and lower case.

In general Notebook Lab did not do exactly what I wanted, which was showing and explaining the difference and improvements I made to the code. However, I never gave the AI the code we had before which might be a huge thing.